

Lonely Older Adults in the Era of Social Media: A Meta-Analytic Review

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Abstract

Objectives: In an increasingly digital modern society, the loneliness of older adults is a pressing public health concern, and social media is considered a potential solution. Despite past reviews that have attempted to synthesize the association between social media usage (SMU) and loneliness in older adults, the precise connection between the 2 remains unclear. The purpose of this study was to quantify the direct relationship between SMU and loneliness in older adults.

Methods: As of August 2023, 3 databases (Web of Science, ProQuest, PubMed) were used for the literature search. A study was included if it measured the relationship between loneliness and SMU in older people over 50 years old. In total, the present study identified 29 effect sizes, representing data from 19 distinct research reports and over 24,877 participants.

Results: The meta-analysis applying a random model, shows a weak negative correlation between SMU and loneliness ($r = -0.06$). The region development moderated the relationship, and specifically, the negative correlation between SMU and loneliness increased to a medium size ($r = -0.24$) when the samples were in developing regions.

Discussion: SMU is negatively correlated with loneliness in old age, which suggests that promotion of SMU among older adults should be implemented, with attention to the inequality between regions and the privacy and availability concerns of older adults. Future research needs to make more efforts in terms of terminology consistency and potential moderators to obtain a more robust understanding of the association between SMU and the loneliness of older adults.

Keywords: Loneliness, Meta-analysis, Region development, Stimulation hypothesis

According to the [World Health Organization \(WHO, 2022\)](#), the number of older adults was just over 1 billion in 2020, which is projected to rise to 2.1 billion by 2050. Such an acceleration of aging is unprecedented and will intensify in the coming years. Faced with declining capacity and progressive states of physical health, loneliness becomes a common experience in later lives, and the prevalence of loneliness has been estimated to be 28.5% among older people, even in high-income countries ([Chawla et al., 2021](#)). Numerous previous studies have provided consistent evidence about the link between loneliness and a series of adverse consequences for older adults, including physical diseases (e.g., cardiovascular disease, stroke, and mortality; [Hu et al., 2021](#); [Perissinotto et al., 2012](#)) and mental health problems (e.g., [Domènech-Abella et al., 2019](#)).

With the advent of the social media era, more older adults have integrated the Internet and social media into their

daily lives. As a key information communication technology (ICT), social media offers the potential to combat loneliness by enabling individuals to connect with others regardless of physical or geographical limitations. Research investigating the impact of social media usage (SMU) on loneliness has expanded over the past two decades, sparking a debate between two opposing perspectives. Proponents of the “displacement hypothesis” argued that replacing offline time (i.e., time with friends and family) with online time decreases the quality of relationships, so SMU increases loneliness ([Valkenburg & Peter, 2007](#)). There exist some warnings targeting older adults against problematic use of the Internet and social media, as people will likely seek refuge in cyberspace and reduce their contact with others, potentially increasing loneliness ([Hunt et al., 2018](#); [Kim et al., 2009](#)). For example, [Sakurai et al. \(2021\)](#) found that older adults who frequently post on Twitter were prone to experience loneliness. Whereas,

the other side advocated the “stimulation hypothesis”: older adults can maintain or even expand their contact with the outside society through SMU, thus relieving their loneliness (Szabo et al., 2019; Wetzel et al., 2021). The COVID-19 pandemic has further underscored the role of ICTs in addressing loneliness. A systematic review analyzed 17 loneliness interventions in old age, and three of the four successful projects involved new technologies (Hagan et al., 2014). Another systematic review also highlighted the essential role of SMU, reporting a shift in development trends in ICT interventions designed for older adults from animal robots to online social platforms (Choi & Lee, 2021). So far, the empirical evidence on the risks and benefits of SMU remains inconclusive, especially for older adults.

Efforts have been made to synthesize the association between SMU and loneliness in older adults. For instance, Wiwatkunupakarn et al. (2022) systematically reviewed the relationship of social networking sites (SNS) with social isolation, loneliness, and depression in old age and found some supportive evidence between SNS and loneliness but with limited experimental studies; Shah et al. (2021) conducted a meta-analysis and suggested no evidence supporting the effectiveness of digital technology interventions for loneliness of older adults; however, another more recent meta-analysis showed that remote intervention could reduce loneliness in older adults, which was affected by media type, treatment strategy, and other factors (Fu et al., 2022). Despite these valuable insights, notable gaps remain. Some reviews only provided qualitative summaries with limited quantitative evidence on the association between SMU and loneliness, while others focused on broader categories of digital or remote interventions, which include but are not limited to SMU. Therefore, a comprehensive and quantitative lack of evidence persists to date, specifically focusing on the direct association between SMU and loneliness among older adults. To fill this gap, we use robust meta-analysis methods to systematically synthesize existing literature to discern the nuanced relation between SMU and loneliness among individuals over 50 and examine the potential moderators that may amplify or shrink this association. Accordingly, we propose an *a priori* hypothesis regarding the overall effect:

Hypothesis: Higher SMU is associated with lower levels of loneliness in older adults.

Despite some studies exploring the relation between SMU and loneliness may differ depending on different variables (e.g., active and passive SMU: S. Yang et al., 2021a), their conclusions remain inconsistent. Therefore, we do not propose specific prior assumptions about the moderating variables, but adopt an exploratory approach to identify potential moderators.

Data and Methods

This systematic review and meta-analysis follow the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines (Page et al., 2021). This study has been registered in PROSPERO (CRD42024533574). The identification of relevant abstracts, selection of studies, subsequent data extraction, and quality assessment were performed independently by two of the authors (R. C. and K. Z.), and conflicts were resolved by a third investigator (S. C.).

Search Strategy

As of August 2023, three electronic databases (Web of Science, ProQuest, PubMed) were searched with no time limitation,

using the following terms: older adult OR aged OR retirement OR old aged OR geriatrics AND (social media OR SNS OR online social networking OR Internet OR Facebook OR Twitter) AND (loneliness OR lonely).

Inclusion Criteria

Studies were included if they: (a) targeted older adults aged over 50; (b) quantitatively examined the relationship between SMU and loneliness; (c) were published in English; (d) reported the correlation coefficients (r) as an effect size indicator, or provided other statistical information (e.g., t , F , χ^2 , β values) that could be converted to r .

The definition of old age

When the WHO (2021) refers to aging, it considers adults over the age of 60, which is a common definition of old age as well. However, there is great variation in how different regions define older age: the retirement ages in many developed Western countries are above 65, whereas the retirement ages in some non-Western countries, like Turkey, are as low as 49 (OECD, 2023). Furthermore, the old age range has been widened to over 50 in many clinical studies (Negin et al., 2012; Sabharwal et al., 2015). Due to the limited number of studies in this field found in the prior literature browsing, we set the age of participants as above 50 to ensure the inclusion of extensive studies.

The range of SMU

As one of the high-profile interventions for loneliness, ICT includes social media. A large body of research on loneliness involves various ICTs. As a consequence, we limit social media to a variety of apps for sharing and communication, including classic social media apps like Facebook, Twitter, WeChat, and Instagram. Since we focused on the impact of social media itself on the psychology of older populations, any studies that included other electronic technologies or Internet use, such as phone calls, E-mail, and online games, or simply used social media as a platform to implement remote psychological interventions (e.g., online mindfulness; Ungar et al., 2019) were not included in the review.

Focus on loneliness

A large number of studies take social isolation and loneliness into account together (Custodio et al., 2022; Meshi et al., 2020), and meanwhile, the two concepts are very confusing. Loneliness is viewed as the subjective experience of negative feelings about the loss of relationships. Social isolation, on the other hand, is connected with the objective characteristics of a situation and refers to the absence of relationships with other people. The corollary between the two is not always absolute (De Jong Gierveld & Havens, 2004). This review concentrated only on loneliness (subjective perception) in older adults and excluded studies that only measured social isolation.

Data Extraction and Moderator Coding

We extracted data on authorship and effect sizes of the association between SMU and loneliness. To investigate key moderators that may amplify or reduce the association between SMU and loneliness among older adults, potential moderators were extracted as well and coded. These details were divided into characteristics related to the (a) study, (b) sample, (c) social media, and (d) loneliness. For those studies where a given moderator was not relevant to the design and/or not

possible to code, they were coded as “not specified” and omitted from the analysis of that specific moderator.

Characteristics related to the study

First, the **publication year** (quantitative: ranging from 2015 to 2023) of the articles was documented. This factor was considered to assess if recent studies, possibly subject to stricter evidence standards, showed smaller or bigger effect sizes compared to older studies (Y. Yang et al., 2023).

Second, we coded studies according to the **type of study design** (qualitative: cross-sectional, experimental, longitudinal). The inclusion of this moderator aims to investigate the potential impact of different study designs on the effect sizes. Since the type of study design is intrinsically linked to the methods of data collection and analysis, it can significantly influence the interpretation and validity of the findings.

Third, the **pandemic timing** (qualitative: before COVID-19, after COVID-19) of studies was coded based on the year they were conducted (or the publication year if the conduct year was not specified in the articles). This moderating factor assesses whether the study was conducted before the COVID-19 pandemic to determine the potential impact of the pandemic on the research outcomes.

Characteristics related to the sample

First, the **sample size of participants** (quantitative: ranging from 36 to 4,936) was recorded. This moderator was incorporated to determine the relationship between sample sizes and their corresponding effect sizes.

Second, the **mean age** (quantitative: ranging from 64.3 to 81.8 years) of all participants in the study was recorded. Research indicates that older adults in their younger years often feel less lonely compared to those in their later years (e.g., F. Yang et al., 2018). Consequently, mean age was considered a potential moderating factor to determine whether certain age groups among older adults showed more significant effects compared to others.

Third, the **proportion of female participants** (i.e., gender composition; quantitative: ranging from 40.4% to 70.2% female) was determined from the demographic data provided for the entire participant group. When participants were excluded and the demographics of the remaining usable sample were reported, these figures were utilized. The majority of studies categorized participants using binary terms of male or female and did not specify if participants reported their sex or their gender. While we use the term “gender” to describe participants, we recognize the limitations of this method in accurately representing individuals’ gender identities. Previous research indicates that older male participants may be less engaged in SMU compared to their female counterparts (e.g., Newman et al., 2021), potentially influencing feelings of loneliness. Therefore, this moderator was included to assess if older male or female participants experienced different levels of loneliness associated with SMU.

Fourth, we coded **region development** (qualitative: developed, developing), according to the Human Development Index (HDI; United Nations Development Programme, 2022). Regions that score over 0.800 are classified as developed, and regions whose HDI score falls below that are classified as developing. Previous research in psychology has established a link between socioeconomic status (SES) disadvantages and loneliness in older adults (e.g., McCain et al., 2021), and the SES decides their digital divide as well (Yoon et al., 2020).

To address the lack of SES environment or relevant data in the majority of included studies, we utilized the development degree of regions where participants were located, allowing us to explore whether the socioeconomic environment could influence the correlation size of SMU and loneliness in older adults.

Characteristics related to social media

First, we coded four specific **SMU outcomes** (qualitative: status, frequency, intensity, interaction style), based on their measurement content reported in the study. Although these studies all looked at older people’s SMU, each study focused on different things. Due to the complexity of social media measurements, the review summarized researchers’ inspection of SMU into four dimensions: (a) status (i.e., whether participants use social media or not); (b) frequency; (c) intensity (e.g., the feeling of involvement in social media, the number of activities participants engaged in, or the number of WhatsApp groups); (d) interaction style (e.g., active and passive). To conduct moderator analyses on the complete data, the correlations of passive SMU were reverse coded.

Second, we summarized and recorded the **strategies for using social media** (i.e., SMU strategy; qualitative: directed communication, open interaction, general use), according to the description in each study. Building upon an intricate understanding of social media’s impact on the mental well-being of older adults, previous studies made a series of useful explorations and attempts. Bonsaksen et al. (2023) suggested that older adults who use social media to maintain relationships may feel lonelier than their peers. Distinctions in social media interactions—active versus passive—appear to have divergent effects on mental health (e.g., Yang et al., 2021). Burke et al. (2011) and Hutto et al. (2015) categorized social media activities into directed communication with specific individuals, broadcast communications to wider audiences, and passive consumption of others’ content. Following the findings above, in our research, we synthesized these activities into three overarching SMU strategies: (a) directed communication, encompassing personal, targeted exchanges; (b) open interaction, reflecting broader, nontargeted communication or passive activities (e.g., browsing or viewing content); and (c) general use, referring to unspecified or mixed social media engagement (e.g., “How much do you use social media on average per day?”). To ensure consistency in categorization, we applied the coding rules based on platform characteristics and usage descriptions: if a study specified a platform, we classified it accordingly (e.g., WhatsApp as directed communication, Facebook as open interaction); if no platform or multiple types was mentioned, it was coded as general use; if studies specified a function (e.g., “video calls” for directed communication, “browsing posts” for open interaction), we categorized based on these descriptions. Different strategies may lead to varying experiences and levels of social support obtained through SMU. Therefore, considering these strategies is essential for a comprehensive understanding of how SMU affects loneliness among older adults.

Characteristics related to loneliness

Four categories of **loneliness measurement** (qualitative: UCLA, ESLS, T-ILS, a single question) were coded in the current study: UCLA (The University of California, Los Angeles Loneliness Scale), ESLS (The De Jong Gierveld short scales

for Emotional and Social Loneliness), T-ILS (the Three-Item Loneliness Scale), and a single question.

Statistical Analysis

All analyses were conducted in R Version 4.2.1. Heterogeneity was computed by τ^2 and interpreted using I^2 statistics. Due to the heterogeneity of the included studies, the meta-analysis was performed by applying a random effect model with the *meta* package. The present meta-analysis concerns the association between SMU and loneliness, and selected Pearson's correlation coefficient as the effect size (first transformed to Fisher's z for analyzing and finally back-transformed to r for the reporting of results). In our analysis, all the Pearson's r were interpreted following the guidelines proposed by [Funder and Ozer \(2019\)](#): $r = 0.05$ suggests minimal impact initially but potential consequence over time; $r = 0.10$ indicates a still small effect that potentially more ultimately consequential; $r = 0.20$ is medium and practical in the short run; $r = 0.30$ is large and powerful both short and long-term; an effect size exceeding $r = 0.40$ is often an exaggerated estimate in psychological studies.

However, due to the inconsistency in reporting numerical findings of this association, the other effect sizes (e.g., standardized mean difference and odds ratio) were transformed into Pearson's correlation coefficients using the website Psychometrica ([Lenhard & Lenhard, 2022](#)). For assessing potential moderators, meta-regression analyses and subgroup analyses were conducted. Additionally, we assessed the robustness of our findings through sensitivity analysis and publication bias assessment.

Results

Characteristics of Included Studies

As shown in [Figure 1](#), 1,287 articles were identified from the three databases. Through screening the title and abstract of the article, 160 articles were left. Reviewers evaluated these articles in full text according to the inclusion criteria, and for various exclusion reasons, 19 articles were left. The final search results are displayed in [Supplementary Material Section 1](#).

As no study was rated as high risk in quality assessments (see [Supplementary Material Section 2](#)), the 19 articles (with a total of 29 samples, as some of the research recruited several samples or used two or more instruments to measure SMU and loneliness) were included in the meta-analysis. The total sample included 24,877 participants (Female % = 52.83%; mean age = 72.17; range: 64.30–81.80). Details of each study are displayed in [Table 1](#).

SMU definitions and measurements

Researchers have approached the definition of SMU through various lenses. They have employed: (a) comprehensive terminology: some researchers (e.g., [Mohan & Lyons, 2022](#)) have employed comprehensive terms such as “social media” or “SNS,” often supplemented by illustrative examples to enhance conceptual clarity; (b) specific platforms: some studies (e.g., [Hutto et al., 2015](#)) focus on distinct social media platforms such as Twitter or Facebook; (c) functional perspectives: studies have also delved into the functional aspects of social media, investigating its roles in videoconferencing ([Tsai et al., 2020](#)), information dissemination ([Chen & Gao, 2022](#)), and so on; (d) usage styles: understanding SMU has

also involved delineating between active and passive styles ([Yang et al., 2021](#)).

SMU was measured in a great variety of ways. Three experimental studies conducted interventions on SMU, which naturally generate a binary variable without measuring. Among 16 observational studies, half of the studies used merely a single or two questions to ask participants about their SMU status or frequency ($n = 8$), five studies utilized validated scales (e.g., Facebook intensity scale), and two applied various self-compiled scales with several items. There remains one study in which SMU was one option for questions on Internet activities ([Mohan & Lyons, 2022](#)).

Loneliness measurements

Over half of the included articles ($n = 12$) applied different versions of The University of California, Los Angeles (UCLA) Loneliness Scale. Four studies utilized The De Jong Gierveld short scales for emotional and social loneliness to measure both dimensions ($n = 2$) or only social loneliness ($n = 2$). Two studies used a three-item loneliness scale specifically for use in large surveys (2004). Finally, one study used a single item to assess loneliness ($n = 1$; [Ho et al., 2023](#)), “How lonely have you felt in recent days?”

Association Between SMU and Loneliness

The heterogeneity test showed that $I^2 = 89.1\%$ and $\tau^2 = 0.014$ ($p < .0001$). The I^2 over 75% indicates high heterogeneity ([Higgins et al., 2003](#)), which suggests that our meta-analysis should apply random-effect models. Meta-analytic results of 29 samples show a weak but significant negative correlation between SMU and loneliness ($r = -0.063$; 95% CI $[-0.114, -0.012]$; $t = -2.56$; $p = .016$; [Figure 2](#)).

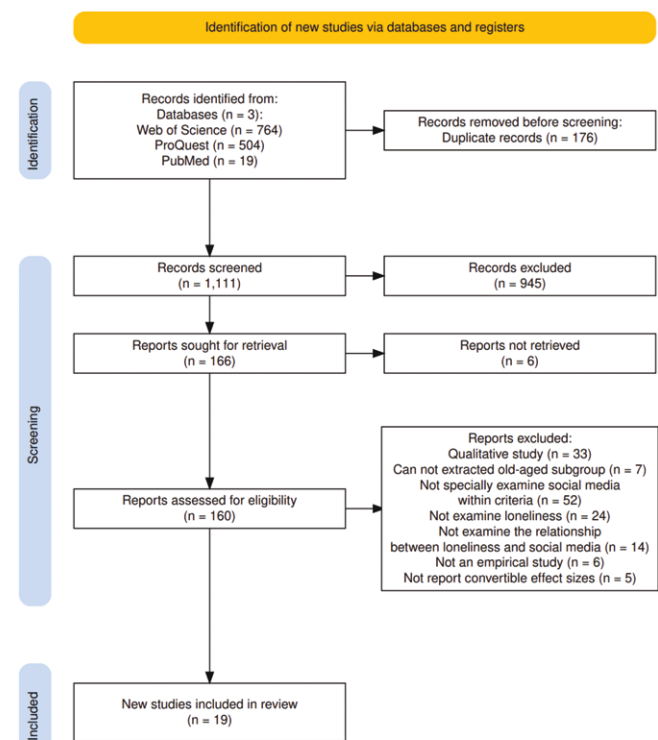


Figure 1. PRISMA flow diagram.

Table 1. Main Characteristics of the Included Studies

Study	K	N	Region	Design	SMU definition	SMU measurement	Loneliness measurement	Quality assessment
Byrne et al. (2021)	1	4,315	United States	Cross-sectional	Social technique use	A question about frequency: "On average, how often do you communicate by Skype, Facebook, or other social media with any of your (children, other family members, friends) not counting any who live with you?"	UCLA—11 items	Low risk
Mohan & Lyons (2022)	1	3,416	Ireland	Cross-sectional	One of types of Internet uses	An option "using social networking sites (e.g. Facebook, Twitter, Myspace)" for the question: "For which of the following activities did you use the Internet in the last 3 months?"	UCLA—5 items	Low risk
Zhou (2018)	1	596	China	Cross-sectional	SNS usage	A self-report scale with five questions, such as "I frequently participate in the activities in this site (referring to the six forums used for participants recruitment such as Baidu Tieba etc.)"	UCLA—11 items	Low risk
Bonsaksen et al. (2021)	4	826	Norway, United States, UK, Australia	Cross-sectional	10 social media channels: Facebook, YouTube, Snapchat, Instagram, LinkedIn, Twitter, Pinterest, WhatsApp, Reddit, and Tumblr	A question about the number of the 10 social media used, and a question about how often participants had used social media in general after the COVID-19 outbreak	ESLS	Low risk
Rolandi et al. (2020)	2	130	Italy	Experimental	SNSs use	Invention: an SNS course consisting of five interactive group sessions on smartphone use, Facebook and WhatsApp use, privacy rules, and fraud risk prevention using Facebook.	UCLA—3 items	Low risk
Quinn (2021)	1	36	United States	Experimental	Facebook and Twitter use	Invention: social media intervention workshop on Facebook and Twitter use	UCLA—11 items	Low risk
Simons et al. (2022)	2	349	Netherlands	Cross-sectional	SNS use and WhatsApp use	An adapted version of the Facebook Intensity Scale for any social network site used by respondents	ESLS	Low risk
van Ingen et al. (2017)	1	2,162	Netherlands	Longitudinal	SNS use	Two questions about status and frequency: "Respondents were asked whether they ever spent time on a list of Internet activities (yes or no), and subsequently how many hours per week they usually spent on these activities."	ESLS—social	Some concerns
Yang et al. (2021)	2	221	China	Cross-sectional	ASMU (active social media use) and PSMU (passive social media use)	A validated scale with seven items, including different social media activities	UCLA—3 items	Low risk
Tsai et al. (2020)	1	62	Taiwan (China)	Experimental	Line-based videoconferencing	Invention: line-based videoconferencing with families at least once a week	UCLA—10 items	Some concerns
Gazit et al. (2022)	2	427	Israel	Cross-sectional	WhatsApp	Two questions about WhatsApp using status and frequency	UCLA—3 items	Low risk
Stockwell et al. (2020)	1	4,492	UK	Cross-sectional	Social networking sites (Facebook, Twitter, Myspace)	One option of questions on Internet activities in the last three months	UCLA—3 items	Low risk
Hutto et al. (2015)	3	141	United States	Cross-sectional	Facebook communications	A validated scale of frequency of Facebook communication with family and friends and typical SNS activities	UCLA—20 items	Low risk
Yang et al. (2021)	1	684	Hongkong (China)	Cross-sectional	Platforms for communicating with one another, such as Facebook, Twitter, WhatsApp, or WeChat	Two questions about status and frequency: The participants were asked whether they had used an SNS in the past 12 months. Those who said "yes" to the question were further asked how many hours per day they had spent, on average, on these SNSs during the COVID-19 pandemic.	ESLS—social	Some concerns

Table 1. Continued

Study	K	N	Region	Design	SMU definition	SMU measurement	Loneliness measurement	Quality assessment
Yu et al. (2015)	2	607	United States	Cross-sectional	Social networks such as Facebook or Twitter	A question about SMU status: "Do you use social networks such as Facebook or Twitter?"	UCLA—11 items	Low risk
Chen & Gao (2022)	1	276	China	Cross-sectional	Informational use of social media (mostly WeChat)	A question about the frequency of informational use: "How often do you interact with the above content on social media (e.g., WeChat)?"	T-ILS	Some concerns
Ho et al. (2023)	1	647	Taiwan (China)	Cross-sectional	Social networking services and instant messaging applications	Two questions about status and frequency: Multiple-choice items on the social media platforms used in the past month, and a question about time spent on LINE and Facebook.	A single question	Low risk
Kim & Lee (2023)	1	554	South Korea	Cross-sectional	Social media such as Twitter, Instagram, Facebook, and TikTok	A question about the frequency: "How much do you use social media on average per day?"	UCLA—6 items	Low risk
Lu et al. (2023)	1	4,936	Canada	Longitudinal	Online social participation (OSP)	A self-completed scale with four items	T-ILS	Low risk

Notes: K = number of effect sizes; N = number of participants; SMU = social media usage; SNS = social networking sites. For Loneliness scale: UCLA = The University of California, Los Angeles (UCLA) Loneliness Scale; ESLS = The De Jong Gierveld short scales for Emotional and Social Loneliness; T-ILS = the three-item Loneliness Scale.

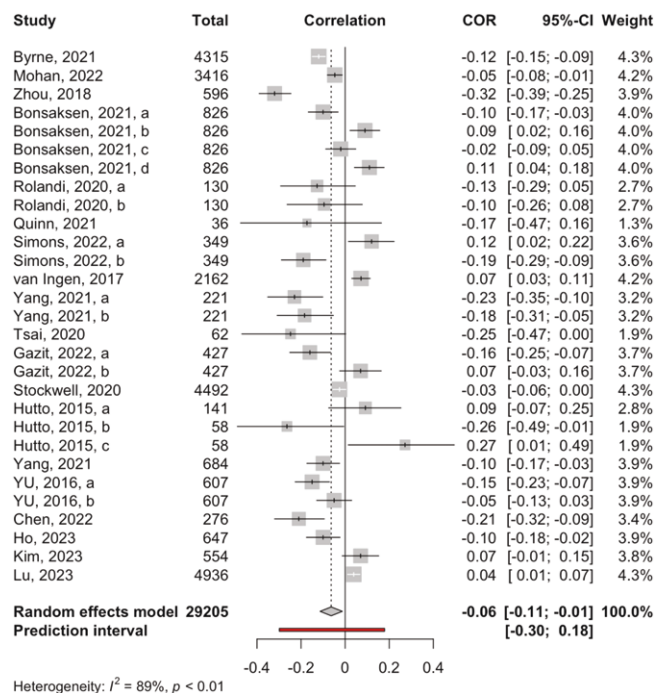


Figure 2. Forest plot for the correlation between social media usage and loneliness (95% CI).

Additionally, the results of the sensitivity analysis and publication bias suggested the robustness of the results (see [Supplementary Material Section 3](#)).

Moderator analyses

Analyses targeted characteristics related to the (a) study, (b) sample, (c) SMU, and (d) loneliness, and results are presented in [Table 2](#) (see [Supplementary Material Section 4](#) for the coding results). Only the region development significantly moderated the overall relation between SMU and loneliness in old age ($F(1, 27) = 12.819$, $p = .001$). To be specific, the negative correlation between SMU and loneliness increased to medium size when the samples were in developing regions (mean $r = -0.242$, 95% CI $[-0.339, -0.141]$), while the effect size of those samples in developed regions was smaller and nonsignificant (mean $r = -0.033$, 95% CI $[-0.080, 0.014]$).

Other factors were not identified as significant moderators in the current study (p values $> .05$ in F tests). Still, mean effect sizes in some specific subgroups were significant: cross-sectional (mean $r = -0.066$, 95% CI $[-0.125, -0.007]$) and experimental (mean $r = -0.177$, 95% CI $[-0.331, -0.013]$) subgroups; SMU status subgroup (mean $r = -0.090$, 95% CI $[-0.140, -0.040]$); UCLA subgroup (mean $r = -0.074$, 95% CI $[-0.141, -0.011]$).

Discussion

Recent public health statistics highlight the vulnerability of older adults to loneliness, prompting the exploration of social media as a potential intervention tool. Some psychology research that has been conducted over the last decade investigating the potential relief of SMU on loneliness in old age, has revealed a nuanced and intricate landscape that consists of results that both support and fail to find evidence for this effect. In this present meta-analysis, we undertook a systematic examination of the relationship between SMU and

Table 2. Results of Categorical and Continuous Moderators for the Association Between SMU and Loneliness of Older Adults

Moderators	<i>k</i>	β/r	95% CI [lower, upper]	<i>F</i> (df1, df2)	<i>p</i>
<i>Study characteristics</i>					
Publication year	29	0.000	[-0.022, 0.022]	<i>F</i> (1, 27) = 0.000	.995
Study design				<i>F</i> (2, 26) = 1.289	.293
Cross-sectional	23	-0.066	[-0.125, -0.007]		
Experimental	3	-0.177	[-0.331, -0.013]		
Longitudinal	3	0.019	[-0.181, 0.217]		
Pandemic timing				<i>F</i> (1, 27) = 0.004	.950
Before COVID-19	13	-0.062	[-0.147, 0.025]		
After COVID-19	16	-0.065	[-0.134, 0.005]		
<i>Sample characteristics</i>					
Sample size	29	0.000	[-0.000, 0.000]	<i>F</i> (1, 27) = 0.896	.352
Mean age	17	-0.002	[-0.015, 0.011]	<i>F</i> (1, 15) = 0.108	.747
Gender composition	22	-0.003	[-0.009, 0.004]	<i>F</i> (1, 20) = 0.686	.417
Region development				<i>F</i> (1, 27) = 12.819	.001
Developed	25	-0.033	[-0.080, 0.014]		
Developing	4	-0.242	[-0.339, -0.141]		
<i>Social media characteristics</i>					
SMU outcomes				<i>F</i> (3, 25) = 0.718	.550
Frequency	8	-0.027	[-0.130, 0.077]		
Status	12	-0.090	[-0.140, -0.040]		
Intensity	5	-0.033	[-0.262, 0.199]		
Interaction style	4	-0.131	[-0.456, 0.225]		
SMU strategy				<i>F</i> (2, 26) = 0.743	.486
Directed communication	9	-0.103	[-0.193, -0.011]		
Open interaction	5	-0.084	[-0.324, 0.166]		
General use	15	-0.036	[-0.103, 0.030]		
<i>Loneliness characteristics</i>					
Loneliness scale				<i>F</i> (3, 25) = 1.372	.274
UCLA	17	-0.076	[-0.141, -0.011]		
ESLS	3	-0.158	[-0.568, 0.315]		
T-ILS	8	-0.001	[-0.098, 0.096]		
A single question	1	-0.100	[-0.324, 0.138]		

Notes: β = estimated regression coefficient; r = mean effect size expressed as a Pearson's correlation; CI = confidence interval; SMU = social media usage. For Loneliness scale: UCLA = The University of California, Los Angeles (UCLA) Loneliness Scale; ESLS = The De Jong Gierveld short scales for Emotional and Social Loneliness; T-ILS = the three-Item Loneliness Scale. Bold text indicates $p < .05$ significance. Pandemic timing refers to whether the study was conducted before or after the COVID-19 pandemic. Gender composition represents the percentage of female participants. Region development indicates whether the study sample was from a developed or developing region. SMU outcomes refer to the dimensions of SMU measured in the study. SMU strategy refers to the specific activities or purposes of SMU.

loneliness of older adults and analyzed the potential moderators. Analyzing 29 effect sizes from 19 studies on 24,877 participants, we found that SMU was weakly related to lower loneliness among older adults. Moderator analyses revealed that the size of the negative correlation between SMU and loneliness among older adults would increase when participants are from developing regions. These findings filled the gap of a quantitative systematic review focusing on the association between SMU and loneliness among older people.

Association Between SMU and Loneliness Among Older Adults

Overall, a significant weak negative correlation was found between SMU and loneliness in old age. In other words, older adults who were more involved with social media were less likely to feel lonely, despite the relatively minimal strength

of this association. The findings offered further support for the "stimulation hypothesis," suggesting that older individuals could leverage social media to maintain or even broaden their connections with society, thereby mitigating their inner loneliness.

However, our meta-analytic results did not support the "displacement hypothesis," possibly owing to functional deficits in older people than in other younger groups. In line with the self-determination theory that identifies a basic psychological need for autonomy as a central feature for understanding effective self-regulation and well-being, SMU serves as an Internet resource that older individuals can harness to cope with the challenges of preserving autonomy in the face of declining functional health (Oswald & Wagner, 2023). Essentially, SMU fulfills their need for autonomy, contributing to well-being and alleviating loneliness.

Despite the statistical significance of the overall relation between SMU and loneliness, caution is warranted in interpreting this effect due to its small effect size ($r = -0.06$). According to the explanation by Funder and Ozer (2019), in our analysis, while each individual instance of SMU may have a minimal impact on loneliness, the cumulative effect across a population or over a longer period could be more significant. Also, it's essential to contextualize these effect sizes by comparing them with similar meta-analyses and established norms in the field. For instance, a recent meta-analysis (2022) examining SNS use and loneliness across all age groups found a weak yet significantly positive correlation ($r = 0.05$). It seems that compared with other age groups, the older population receives more benefits from SMU, just as previous discussion based on self-determination theory. However, it's crucial to note that this analysis did not observe a significant moderating effect of age due to the absence of conclusive coding for older adult subgroups. This highlights the need for further research focusing specifically on the older population to understand the nuanced relationship between SMU and loneliness in older age groups.

Evaluation of Moderators

Moderating factors were chosen based on theoretical propositions and prior research, encompassing characteristics related to the study, sample, SMU, and loneliness. Nevertheless, these analyses largely yielded null or inconclusive effects, with the exception of region development as a significant moderator.

Moderating role of region development

The marginal inclination indicating a negative correlation between SMU and loneliness was more pronounced in older adults residing in developing regions, reaching a medium effect size ($r = -0.242$). This moderating effect may be attributed to the socioeconomic variation, that older individuals in developed areas tend to possess greater wealth and engage in more fulfilling offline activities, relying less on social media for mitigating loneliness. Another possible explanation is the higher prevalence of SMU among older adults in developed areas compared to developing ones, contributing to a diminished sense of empowerment from SMU for those in developed areas. It is also crucial to acknowledge the inequities in data availability, as only 4 out of 29 samples originated from developing regions in our analysis. A similar pattern emerges in another meta-analysis encompassing loneliness data from 113 countries, where data on adults is predominantly concentrated in Europe (Surkalim et al., 2022). Developed regions, such as Europe, lead in surveillance with a wealth of data sources, while many developing regions face significant data coverage gaps. These gaps may stem from limited resources and competing priorities, with issues like food security, housing, and basic healthcare provision taking precedence.

Other nonsignificant moderators

The significant effect sizes observed in certain subgroups still deserve to be discussed despite the absence of significant moderating effects.

First, in terms of study design, contrary to the longitudinal subgroup, both cross-sectional and experimental subgroups showed significant mean effect sizes. While cross-sectional and experimental studies offer insights into immediate associations and causal relationships, longitudinal studies delve into the evolving dynamics over time. The nonsignificant

findings in longitudinal studies may underscore that the effects of SMU on loneliness in older adults may require a more extended observation period or consideration of additional factors that unfold over time.

Second, within these studies measuring various outcomes of SMU, only those that focus on SMU status (i.e., whether to use social media) revealed a significant negative relation between SMU and loneliness in old age. More frequent, intense, and active SMU did not correlate with lower loneliness in older adults, but access to social media did contribute to comfort, aligning with the self-determination theory as well. For older people, SMU holds greater significance as a symbolic expression of autonomy and empowerment, transcending its role as a mere tool for alleviating boredom.

Third, although loneliness measurement didn't moderate the association between SMU and loneliness in old age, the correlation was significant in the UCLA subgroup, not in other subgroups (i.e., ESLS, T-ILS, a single question). The reason may be the differences in reliability, validity, and universality. The brevity of T-ILS and a single question may limit their abilities to capture the complexity of loneliness in the context of SMU among older individuals. ESLS deconstructs loneliness into emotional and social dimensions, whose predictors in older adults were found to differ from each other (Dahlberg & McKee, 2014). Therefore, in our meta-analysis, consisting of heterogeneous samples, studies in ESLS subgroup yielded dispersed patterns of correlation of SMU and loneliness.

Limitations

There are certain drawbacks to this meta-analytic review that are supposed to be taken into account when interpreting these findings. First, the number of included articles was relatively small, despite some recent meta-analyses on a similar or even smaller sample size still yielding valuable insights (e.g., Huffman et al., 2021; $k = 16$). There remained some articles inevitably left out, such as articles written in a non-English language or not answered by email. This underscores the need for more academic attention to the association between SMU and loneliness in old age. Second, the range of social media is hard to define. In a large number of articles, variables such as ICT and Internet use cover SMU. We excluded articles that did not specifically measure SMU and its association with loneliness in our screening process, potentially overlooking relevant literature. Third, the predominance of observational studies and the scarcity of experimental studies hinder making causal inferences about the relationship between SMU and loneliness. More experimental studies are expected to enrich the evidence in this field. Last, the moderation effect of region development should avoid assumptions of universality. The majority of included studies took place in developed regions, China being the only developing country, and lacked those of low region development (e.g., some African countries). Such regional distribution was uneven, and this may contribute to a biased understanding of cross-cultural differences in the observation of SMU and loneliness in old age.

Practical Implications

Here, we offer several considerations for future academic efforts and sociopolitical work.

Initially, social media should be encouraged as an intervention tool for loneliness in old age, based on the significant negative association between SMU and loneliness among older

adults found in this meta-analysis. The government should vigorously promote SMU among older adults. However, relevant practitioners should consider the privacy and security issues raised in some included studies (Hutto et al., 2015; Mohan & Lyons, 2022). Specific educational programs or computer clubs are suggested to be designed for older adults to address issues such as fraud, and misinterpretation of data collection. Technical developers are also advised to address usability issues and improve system design, concerning the usability problems caused by age-related impairments (e.g., vision, handshaking, and memory loss; Frik et al., 2019).

Then, it is crucial to establish uniform terminology in future research. Although we clarified the definitions of older adults and SMU in our review, there remained a great heterogeneity of included samples. Adopting and implementing universally accepted definitions can streamline the result comparisons directly, leading to the creation of more universally applicable and understandable conclusions in the long run.

Moreover, some potential moderators deserve more exploration in future research (e.g., SMU outcomes and strategy). The definitions and measurements of SMU in the present meta-analysis did not follow a uniform form, which highlighted the complexity of SMU. The variable should be disassembled and analyzed for its impact on mental health in further geriatric research.

Finally, the equity issue of aging loneliness attention across regions with various development levels should be underscored. As previously mentioned, the negative correlation between SMU and loneliness was more pronounced among older adults living in developing areas, and additionally there indeed exists a scarcity of data from areas of low human development. Therefore, it is imperative to integrate loneliness into overall health surveillance with extensive geographical inclusivity, employing standardized and validated instruments.

Conclusion

This study revealed a significant, yet weak, negative correlation between SMU and loneliness among older adults. Despite the statistical significance, the findings should be interpreted cautiously, due to the small effect size and high heterogeneity between studies. In addition, in old age, the relation between SMU and loneliness would increase to a medium effect size when participants are from developing regions. Other moderator analyses testing relevant factors were not significant, but some of them (e.g., SMU outcomes) exhibit significant mean effect sizes in certain areas, worth exploring in the future. We consolidate these results and present numerous recommendations for future research to enhance the strength and decisiveness of this research domain, and for sociopolitical initiatives to improve the well-being of older populations.

Supplementary Material

Supplementary data are available at *The Journals of Gerontology, Series B: Psychological Sciences and Social Sciences* online.

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Conflict of Interest

None.

Data Availability

This study has been registered in PROSPERO (CRD42024533574). All the data, analytic methods, and study materials are available and can be accessed in Supplementary Material.

Author Contributions

R. Chen, S.-L. Chen, K. Zhang, L. Cui, and N. Feng conceptualized and designed the study. R. Chen and K. Zhang conducted the systematic literature search and performed the title and abstract screening, quality appraisal, and data extraction. RC conducted data management, data analysis, and synthesis of data. S.-L. Chen, L. Cui, and N. Feng supervised and guided the process of literature screening and data analysis. R. Chen prepared the original manuscript, and all authors reviewed and edited the manuscript and approved the final version of the article. L. Cui obtained the funding for the research.

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